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| REV STATUS<br>OF SHEETS | REV   |  |  |  |  |  |  |  |  |  | REVISIONS |     |             |      |          |
|-------------------------|-------|--|--|--|--|--|--|--|--|--|-----------|-----|-------------|------|----------|
|                         | SHEET |  |  |  |  |  |  |  |  |  | ZONE      | REV | DESCRIPTION | DATE | APPROVED |
|                         |       |  |  |  |  |  |  |  |  |  |           |     |             |      |          |
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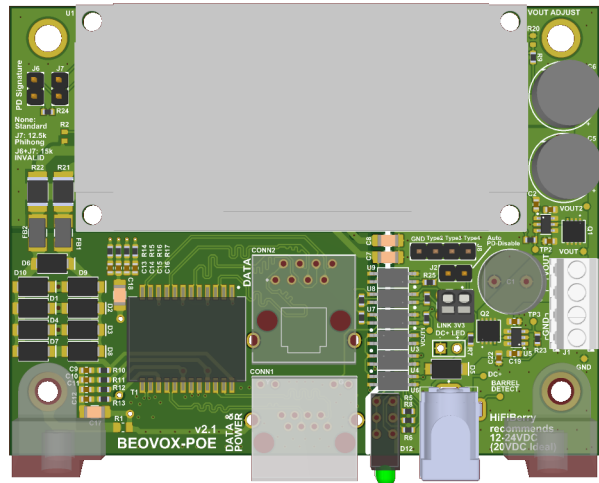
# BEOVOX-POE v2.1

Stackup based on impedance controlled JLCPCB stackup  
JLC041611-7628

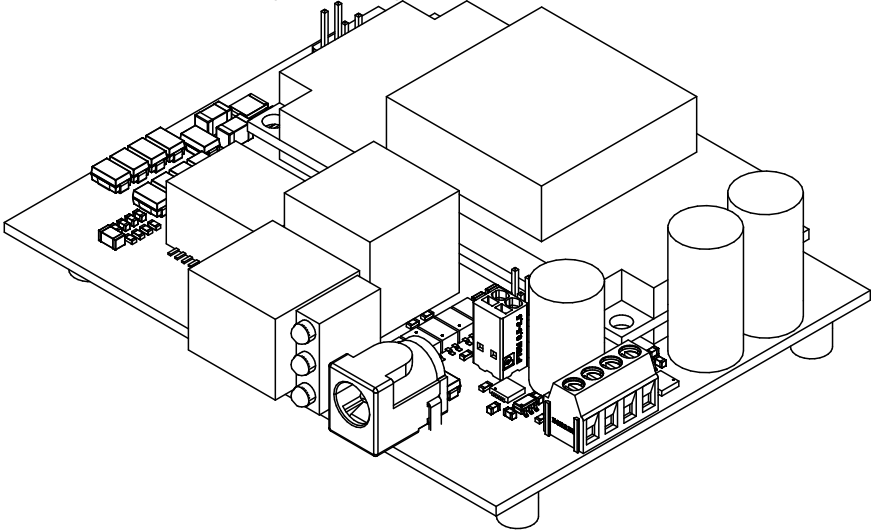
## Layer Stack Legend

| Material                | Layer          | Thickness | Dielectric Material | Type        | Gerber |
|-------------------------|----------------|-----------|---------------------|-------------|--------|
|                         | Top Overlay    |           |                     | Legend      | GTO    |
| Surface Material        | Top Solder     | 0.03mm    | Solder Resist       | Solder Mask | GTS    |
| Copper                  | Top Layer      | 0.04mm    |                     | Signal      | GTL    |
| Prepreg                 |                | 0.20mm    | PP-006              | Dielectric  |        |
| CF-004                  | Layer 2        | 0.04mm    |                     | Signal      | G1     |
| Core                    |                | 1.03mm    | FR-4                | Dielectric  |        |
| CF-004                  | Layer 3        | 0.04mm    |                     | Signal      | G2     |
| Prepreg                 |                | 0.20mm    | PP-006              | Dielectric  |        |
| Copper                  | Bottom Layer   | 0.04mm    |                     | Signal      | GBL    |
| Surface Material        | Bottom Solder  | 0.03mm    | Solder Resist       | Solder Mask | GBS    |
|                         | Bottom Overlay |           |                     | Legend      | GBO    |
| Total thickness: 1.63mm |                |           |                     |             |        |

Realistic View



View from Front side (Scale 1:1)

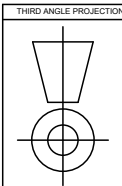


## PCB Configuration

|                   |                 |
|-------------------|-----------------|
| Copper Thickness  | 1oz, all layers |
| Soldermask        | Black or Green  |
| Silkscreen        | White           |
| Surface           | HASL or ENIG    |
| Minimum Clearance | 6mils           |
| Minimum Hole Size | 0.3mm           |
| Blind Vias        | No              |
| Impedance Match   | Recommended     |
| Material          | FR-4            |
| Via Fill          | No              |
| Board Thickness   | 1.6mm           |

Transmission Line Structure Table

| Impedance Id | Transmission Line              | Target Impedance | Calculated Impedance | Trace layer  | Trace Width | Gap     | Reference layers | Target Tolerance |
|--------------|--------------------------------|------------------|----------------------|--------------|-------------|---------|------------------|------------------|
| 1            | Edge-Coupled Coated Microstrip | 100              | 99.54                | Top Layer    | 0.206mm     | 0.190mm | Layer 2          | 10%              |
| 2            | Edge-Coupled Coated Microstrip | 100              | 99.54                | Bottom Layer | 0.206mm     | 0.190mm | Layer 3          | 10%              |

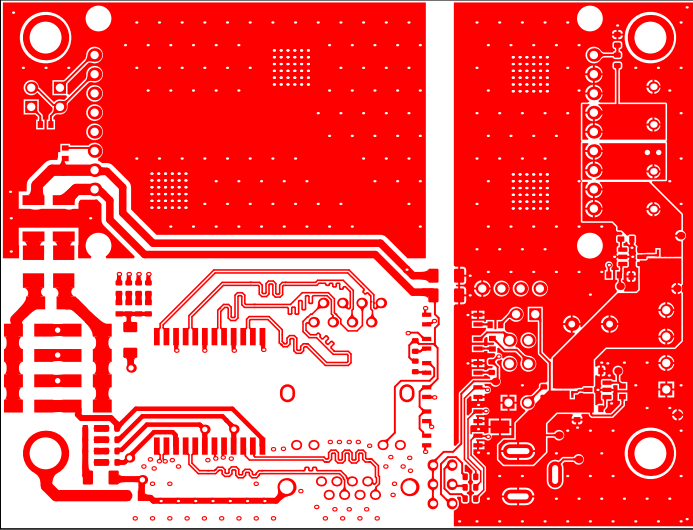


|                           |  |                              |  |                          |  |                      |  |                                                  |  |
|---------------------------|--|------------------------------|--|--------------------------|--|----------------------|--|--------------------------------------------------|--|
| PART NO: =PCB_PART_NUMBER |  | APPROVALS                    |  | DATE                     |  |                      |  | ADDRESS 1<br>ADDRESS 2<br>ADDRESS 3<br>ADDRESS 4 |  |
| ENGINEER: JOSH HORNE      |  | JOSH                         |  | JOSH                     |  |                      |  |                                                  |  |
| DESIGNER: =PCB_DESIGNER   |  | =PCB_DESIGNER                |  |                          |  |                      |  |                                                  |  |
| CHECKER: =PCB_CHECKER     |  | =PCB_CHECKER                 |  | =PCB_CHECKER             |  | DESIGN ITEM: .Item   |  | DESIGN ITEM REVISION: v2.1                       |  |
| BOM DOC: =DOC_NO_BOM      |  | ASSY DOC: =DOC_NO_FAB_DWG    |  | SCH DOC: =DOC_NO_SCH_DWG |  | PCB DOC: =PCB_DWG_NO |  | TITLE: BEOVOX-POE v2.1                           |  |
| SIZE: B                   |  | CAGE CODE: =CAGE_CODE        |  | DWG NO:                  |  | REV:                 |  |                                                  |  |
| SCALE:                    |  | FILE NAME: beovox-poe.PCBDwf |  | SHEET: 1                 |  | OF 2                 |  |                                                  |  |

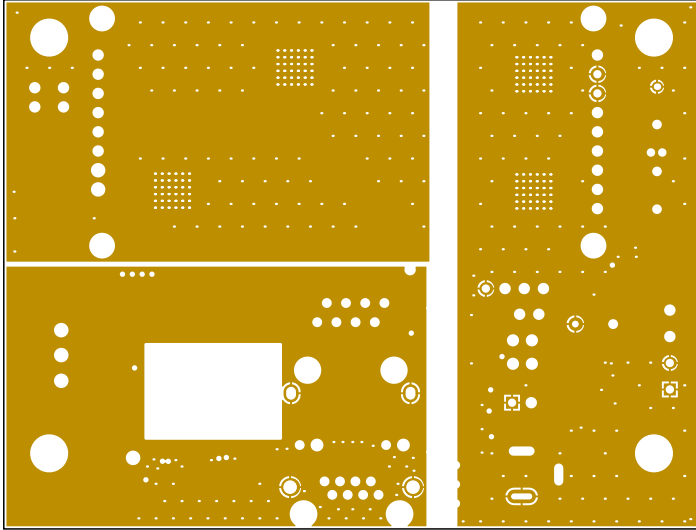
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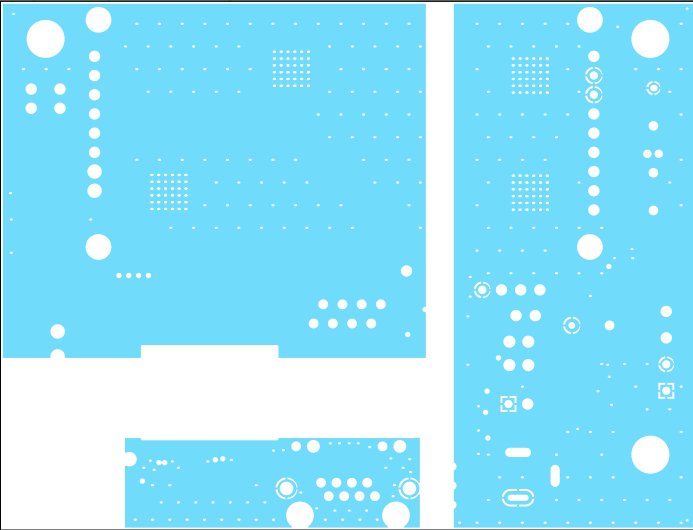
Top Layer (Scale 1:1)



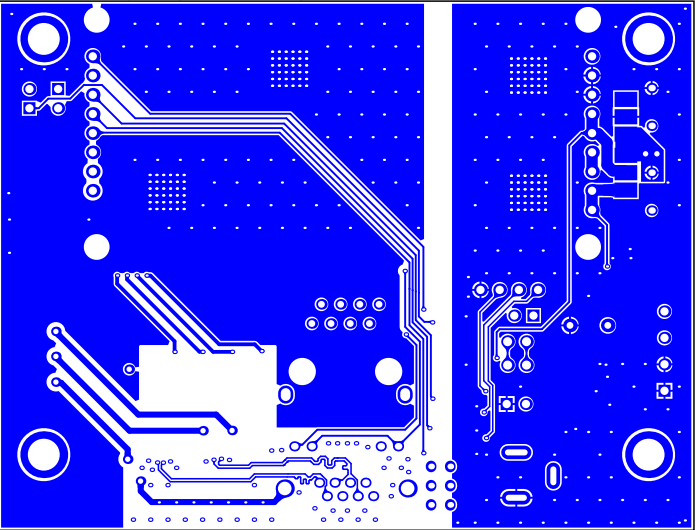
Layer 2 (Scale 1:1)



Layer 3 (Scale 1:1)



Bottom Layer (Scale 1:1)



|                           |  |                      |  |                                      |                                                                                       |                                     |           |                            |          |
|---------------------------|--|----------------------|--|--------------------------------------|---------------------------------------------------------------------------------------|-------------------------------------|-----------|----------------------------|----------|
| PART NO: =PCB_PART_NUMBER |  | APPROVALS            |  | DATE                                 |  |                                     | ADDRESS 1 |                            |          |
| ENGINEER: JOSH HORNE      |  | JOSH                 |  | ADDRESS 2                            |                                                                                       |                                     |           |                            |          |
| DESIGNER: =PCB_DESIGNER   |  | =PCB_DESIGNER        |  | ADDRESS 3                            |                                                                                       |                                     |           |                            |          |
| CHECKER: =PCB_CHECKER     |  | =PCB_CHECKER         |  | =PCB_CHECKER                         |                                                                                       | DESIGN ITEM: .Item                  |           | DESIGN ITEM REVISION: v2.1 | <b>V</b> |
| Reference Documents       |  | BOM DOC: =DOC_NO_BOM |  | ASSY DOC: =DOC_NO_FAB_DWG            |                                                                                       | TITLE: BEOVOX-POE v2.1              |           |                            |          |
| SCH DOC: =DOC_NO_SCH_DWG  |  | PCB DOC: =PCB_DWG_NO |  | SIZE: CAGE CODE: <b>B</b> =CAGE_CODE |                                                                                       | DWG NO: REV:                        |           |                            |          |
| NEXT ASSY                 |  | USED ON              |  | APPLICATION                          |                                                                                       | SCALE: FILE NAME: beovox-poe.PCBDwf |           | SHEET: 2 OF 2              |          |